2019 세포생물학 중간고사

출제하신 교수님 : 김완기

Part 1 (1 point each) <배우지 않은 용어는 무시하시면 됩니다>

1. acid hydrolase **2**. adaptor protein **3**. Arf protein **4**. autophagy **5**. BiP **6**. calnexin **7**. calreticulin **8**. caveola **9**. clathrin **10**. clathrin-coated vesicle

11. constitutive secretory pathway **12**. COPI-coated vesicle **13**. COPII-coated vesicle **14**. cytoplasm **15**. cytosol **16**. ceramide **17**. dolichol **18**. dynamin **19**. ESCRT protein complex **20**. early endosome

21. ER retention signal **22**. ER signal sequence **23**. flippase **24**. free ribosome **25**. glycoprotein **26**. GPI anchor **27**. late endosome **28**. LDL particle **29**. lumen **30**. membrane-bound ribosome

31. microsome **32**. mitochondrial Hsp70 **33**. mitochondrial precursor protein **34**. M6P receptor protein **35**. multivesicular body **36**. N-linked glycosylation **37**. nuclear export receptor **38**. nuclear import receptor **39**. nuclear lamin **40**. NSF

41. O-linked glycosylation **42**. OXA complex **43**. peroxin **44**. phagocytosis **45**. phagosome **46**. phosphatidylinositide **47**. pinocytosis **48**. plasmalogen **49**. Rab effector **50**. Rab protein

51. Ran **52**. Ran-GAP **53**. Ran-GEF **54**. recycling endosome **55**. regulated secretory pathway **56**. rough ER **57**. SAM complex **58**. Sar1 protein **59**. scramblase **60**. Sec61 complex

61. secretory vesicle **62**. signal peptide **63**. signal patch **64**. signal-recognition particle (SRP) **65**. smooth ER **66**. SNARE **67**. SRP receptor **68**. TOM complex **69**. transcytosis **70**. vesicular tubular cluster

Match the definition below with its term from the list above. <u>Caution</u>: number only

1. Contents of the main compartment of the cell, excluding the nucleus and membrane-bounded compartments such as endoplasmic reticulum and mitochondria. ( )

2. Protein sorting signal that consists of a specific three-dimensional arrangement of atoms on the folded protein's surface. ( )

3. Monomeric GTPase present in both cytosol and nucleus that is required for the active transport of macromolecules into and out of the nucleus through nuclear pore complexes. ( )

4. Protein that binds nuclear localization signals from the cytosol into the nucleus through the nuclear pore complexes. ( )

5. Multisubunit protein assembly that transports proteins across mitochondrial outer membrane. ( )

6. Part of a multisubunit protein assembly that is bound to the matrix side of the TIM23 complex and acts as a motor to pull the precursor protein into the matrix space. ( )

7. The most abundant class of phospholipids in myelin, which is synthesized in animal peroxisomes. ( )

8. Type of lipid linkage by which some proteins are bound to the membrane. ( )

9. Short amino acid sequence on a protein that keeps it in the ER. ( )

10. Ribonucleoprotein complex that binds an ER signal sequence on a partially synthesized polypeptide chain and directs the polypeptide and its attached ribosome to the ER. ( )

11. Small vesicle that is derived from fragmented ER produced when cells are homogenized. ( )

12. Hydrophobic amino acid sequence that halts translocation of a polypeptide chain through the ER membrane, thus, anchoring the protein chain in the membrane. ( )

13. A protein that mediates binding between the clathrin coat and transmembrane proteins, including transmembrane cargo receptors. ( )

14. The protein that catalyzes the assembly of the helical domains of paired SNARE proteins. ( )

15. Coated vesicle that transports material from the plasma membrane, and between endosomal and Golgi compartments. ( )

16. Pathway for exocytosis that operates mainly in cells specialized for the secreting products rapidly on demand. ( )

17. Complex vesicle with invaginating buds and internal vesicles involved in the maturation of early endosomes into late endosomes. ( )

18. Invagination that forms from lipid rafts at the cell surface and buds off internally from pinocytotic vesicles. ( )

19. The coat-recruitment GTPase responsible for COPII coat assembly at the ER membrane. ( )

20. Digestion of obsolete parts of the cell by the cell's own lysosome. ( )

21. A family of GTPases that regulate many steps od membrane traffic, including vesicle formation, vesicle movement, and membrane fusion. ( )

22. The uptake of material at one face of a cell by endocytosis, its transfer across a cell in vesicles, and its discharge from another face by exocytosis. ( )

15 63 51 38 68 32 48 26 21 64

31 22 2 40 10 55 35 8 58 4

50 69 54 66 41

23. Membrane-bounded organelle that plays an important part in adjusting the concentration of specific plasma membrane proteins such as glucose transporter. ( )

24. A large protein superfamily whose primary role is to mediate docking of, for example, synaptic vesicles with the presynaptic membrane in neurons. ( )

25. Type of glycosylation that occurs in the Golgi apparatus. ( )

Part II (1 point each)

Underlined amino acid sequences are signal sequences. Predict the intracellular location of proteins containing them. Cytosol (C), Nucleus (N), Mitochondria (M), Endoplasmic reticulum (ER), Lysosome (L), Golgi apparatus (GA), Peroxisome (P).

1. PALISKL-COO^- ( )

2. ^+H_3N-MMSFVSLLLVGILFWATEAENLTKCEVFN----------AAPLIFKDEL-COO^- ( )

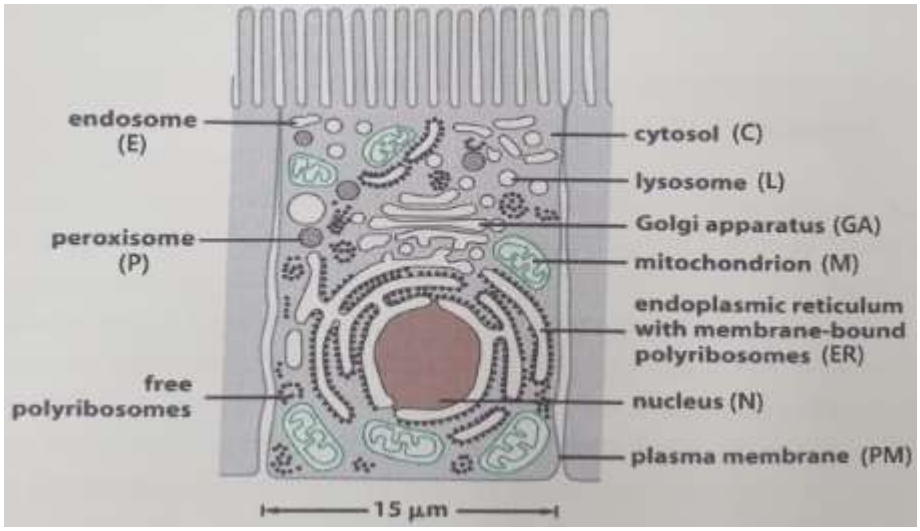
3. ^+H_3N-MMSFVSLLLVGILFWATEAENLTKCEVFN ( )

4. ^+H_3N-MLSLSLRNSIRFFKPATRTLCSSRYLL ( )

5. TNAPPRRTKKPPEFFCAGH ( )

Part III (1 point each)

The major intracellular compartments of a gut epithelial cell are shown below. Specify the location where following proteins function or events occur.



1. acidic phospholipase ( ) 2. GlcNAc phosphotransferase ( ) 3. GPI anchor attachment ( )

4. membrane synthesis ( ) 5. Ran-GAP ( ) 6. scramblase ( )

7. addition of N-acetylgalactosamine to serine or threonine ( ) 8. formation of caveolae ( )

9. dissociation of the LDL receptor from LDL particles ( ) 10. flippase ( ) 11. OXA complex ( )

12. plasmalogen biosynthesis ( )

L GA(G) ER ER C ER GA(G) PM E PM M P

Part IV (1 point each)

Choose <u>eight true</u> statements.

1. The TOM complex is required for the import of all nucleus-encoded mitochondrial proteins.

2. The two signal sequences required for transport of nucleus-encoded proteins into the mitochondrial inner membrane via the TIM23 complex are cleaved off the protein in different mitochondrial compartments.

3. Peroxisomes are found in all types of eukaryotic cells.

4. Due to the many specialized transport proteins in the mitochondrial outer membrane space is chemically equivalent to the cytosol with respect to small molecules.

5. The signal peptide, when it emerges from the ribosome, binds to a hydrophobic site on the

ribosome causing a pause in protein synthesis, which resumes when SRP bind to the signal peptide.

6. Nascent polypeptide chains are transferred across the ER membrane through a pore in the Sec61 protein translocator complex.

7. In multipass transmembrane proteins the odd-numbered transmembrane segment (counting from the N-terminus) act as start-transfer signals and the even-numbered segments act as stop-transfer signals.

8. Some membrane proteins are attached to the cytoplasmic surface of the plasma membrane through a C-terminal linkage to a glycosylphosphatidylinositol (GPI) anchor.

9. In all events involving fusion of a vesicle to a target membrane, the cytosolic leaflets of the vesicle and target bilayers always fuse together, as do the leaflets that are not in contact with the cytosol.

10. Complementary Rab proteins on transport vesicles and target membranes bind to one another to allow transport vesicles to dock selectively at their appropriate target membranes.

11. There is one strict requirement for the exit of a protein from the ER: it must be correctly folded.

12. All of the glycoproteins and glycolipids in intracellular membranes have oligosaccharide chains facing the luminal side, whereas those in the plasma membrane have their oligosaccharide chains facing the outside of the cell.

13. Lysosomal membranes contain a proton pump that utilizes the energy of ATP hydrolysis to pump protons out of the lysosome, thereby maintaining the lumen at a low pH.

14. Late endosomes are converted to mature lysosomes by the loss of distinct endosomal membrane proteins and a further decrease in their internal pH.

15. All the molecules that enter early endosomes ultimately reach late endosomes where they become mixed with newly synthesized acid hydrolases and end up in lysosomes.

16. When a gene encoding a secretory protein is transferred to a secretory cell that normally does not make the protein, the foreign protein is packaged into secretory vesicles.

Answers: ( )( )( )( )( )( )( )

Answer

PART I

15 63 51 38 68 32 48 26 21 64

31 22 2 40 10 55 35 8 58 4

50 69 54 66 41

PART II

P ER ER M C

PART III

L GA(G) ER ER C ER GA(G) PM E PM M P

PART IV

1 3 6 9 11 12 14 16